A Project Proposal Report on

Library Management System

A Project Report Submitted for the Course

Object-Oriented Programming (OOP)

Submitted by:

Sambriddhi Kunwar

Date:

09/08/2026

KING'S COLLEGE
architecting future

King's College Nepal

Bijulibajar, Kathmandu

ABSTRACT

The Library Management System is a console-based application developed using Object-Oriented Programming (OOP) in C++ to manage the basic operations of a library efficiently. The system is designed to allow librarians and users to manage books, search for available books, borrow and return books, and maintain library inventory. The system reduces the need for manual record-keeping and provides an organized approach to managing library resources.

The project applies important OOP concepts such as encapsulation, inheritance, polymorphism, and abstraction to create a modular and maintainable system. Different classes will be developed to represent entities such as books, users, librarians, and library records. File handling will be used to store and retrieve information about books, users, and borrowing transactions.

The main purpose of this project is to demonstrate the practical implementation of OOP concepts while developing a useful library management application. The proposed system will provide basic library functionality and can be extended in the future with database integration, graphical user interfaces, barcode scanning, and online book reservation.

Keywords:*C++, Object-Oriented Programming, Library Management System, Books, Inventory Management, Encapsulation, Inheritance, Polymorphism, Abstraction, File Handling.*

Table of Contents

1. INTRODUCTION

Libraries contain a large number of books and other resources that need to be properly organized and maintained. Traditional methods of managing library records using notebooks or spreadsheets can be time-consuming and may result in errors. A computerized Library Management System can simplify these activities by providing an organized method for managing books, users, borrowing records, and inventory.

This project proposes the development of a Library Management System using Object-Oriented Programming in C++. The system will allow users to search for books, borrow available books, return borrowed books, and view library information. A librarian or administrator will be able to add, update, and remove books and manage the library inventory.

The project will demonstrate important OOP principles, including encapsulation, inheritance, abstraction, and polymorphism. File handling will also be implemented to store information permanently. The system will provide practical experience in software design, programming, data management, and the development of a real-world application using C++.

1.1 PROBLEM STATEMENT

Managing a library manually can be time-consuming and difficult, especially when the number of books and users increases. Traditional methods of maintaining records can lead to errors, misplaced information, and difficulties in tracking the availability of books. Librarians may also face challenges when recording borrowed and returned books, updating inventory, and maintaining user borrowing histories.

Searching for a particular book manually can take additional time and may not always provide accurate information about its availability. Therefore, there is a need for a simple and organized Library Management System that can automate these basic tasks. The proposed system will allow users to search for books, borrow and return books, while librarians can add, update, and delete books and manage the library inventory.

The system will be developed using C++ and Object-Oriented Programming concepts, providing an efficient way to manage library records while demonstrating the practical application of OOP principles.

1.2 PROJECT OBJECTIVES

The main objectives of this project are:

1. To develop a Library Management System using C++.
2. To apply Object-Oriented Programming concepts such as encapsulation, inheritance, polymorphism, and abstraction.
3. To allow librarians to add, update, and remove books.
4. To allow users to search for books.
5. To allow users to borrow available books.
6. To allow users to return borrowed books.
7. To maintain accurate library inventory records.
8. To track borrowed and returned books.
9. To use file handling for persistent data storage.

1.3 SCOPE AND LIMITATIONS

The scopes of this project are:

1. Book Management: Add new books,Update book information,Delete books, Display all books and search books by title, author or category.
2. User Management: Register library users, Store user information, Search user records and View borrowing history.
3. Borrowing and Returning: Borrow available books, Return borrowed books, Update book availability automatically and maintaining borrowing records.
4. Library Inventory: View total number of books, Track available and borrowed books and manage different boom categories.
5. File Management: Store book information, store user information and store borrowing records.

The limitations of this project are:

1. It will use files instead of a database.
2. It will not support online book reservations.
3. It will not include real-time synchronization across multiple computers.
4. Authentication and security features will be basic.

2. LITERATURE REVIEW

Library management systems have become important tools for organizing and managing large collections of books and user records. Traditional manual library systems require librarians to maintain physical records of books, users, and borrowing transactions. Computerized systems can reduce manual work and improve the accuracy and accessibility of library information.

Object-Oriented Programming provides an effective approach for developing management systems because real-world entities can be represented as objects. According to Stroustrup (2013), C++ supports object-oriented programming features that allow developers to create reusable and organized software components. Classes can represent entities such as books, users, and library transactions.

Sommerville (2016) explains that software systems benefit from structured design and modeling because it allows developers to identify system requirements and relationships before implementation. UML diagrams such as use case diagrams, class diagrams, and sequence diagrams can therefore be used to design the proposed Library Management System.

Pressman and Maxim (2020) emphasize the importance of software development processes, system analysis, design, implementation, and testing when developing reliable applications. Applying these principles to a library system can help ensure that individual modules work correctly and interact effectively.

This project combines these concepts by using C++ OOP, UML modeling, and file handling to develop a simple but functional Library Management System.

3. METHODOLOGY

The project will follow the Software Development Life Cycle (SDLC) using a structured development approach.

3.0.1 Requirement Analysis

The requirements of the system will be identified by considering the activities performed by librarians and library users. major requirements include: Book registration, Book searching, borrowing and returning, User management and Inventory management.

3.0.2 System Design

The system structure will be designed using UML diagrams. The diagrams that will be developed are Use Case Diagram, Sequence Diagram and Class Diagram. These diagrams will represent system interactions, processes, classes, and relationships.

3.0.3 Implementation

The system will be implemented using C++. OOP concepts that will be used are Encapsulation which will protect book and user information through private data members. Inheritance which will create specialized classes such as Librarian and Member from a common User class. Polymorphism which allows different user types to perform operations differently when required. Abstraction which hids unnecessary implementation details from users.

3.0.4 Testing and Evaluation

The system will be tested using different scenarios which includes Adding new book, Searching for a book, Borrowing an available book,Attempting to borrow an available book, Returning a book, Updating inventory and Deleting a book.

3.1 USE CASE DIAGRAM

The Use Case Diagram will show how the main actors interact with the Library Management System.

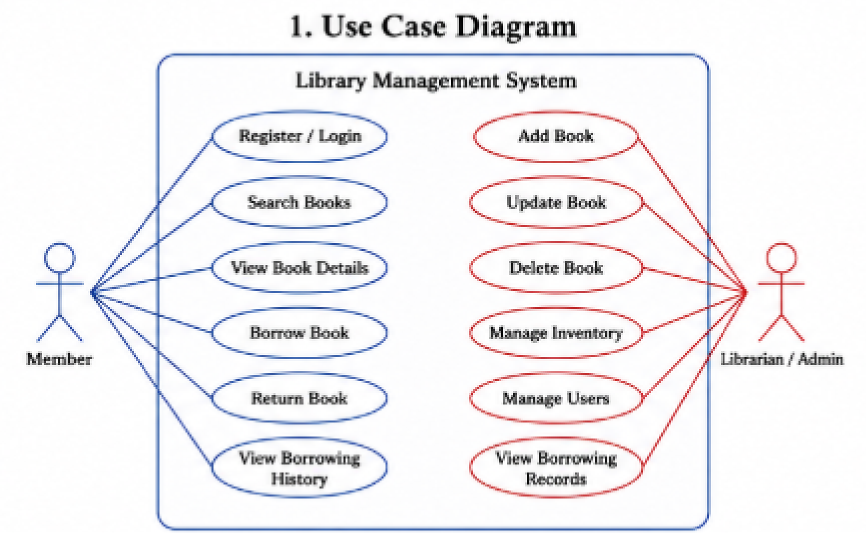

Figure 1: Use Case Diagram

3.2 SEQUENCE DIAGRAM

The Sequence Diagram will demonstrate the process of borrowing a book.

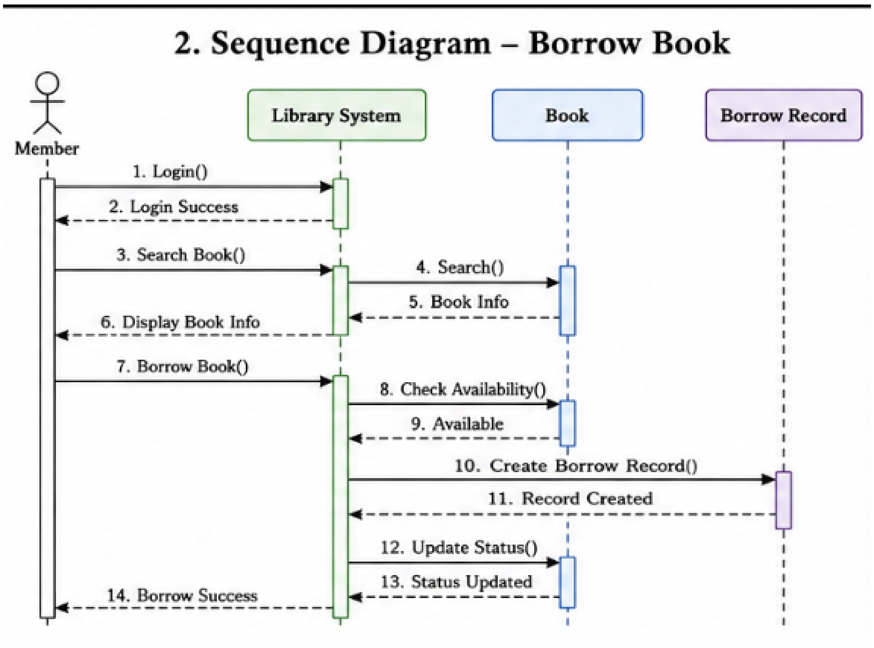

Figure 2: Sequence Diagram

3.3 CLASS DIAGRAM

The proposed system will contain several major classes.

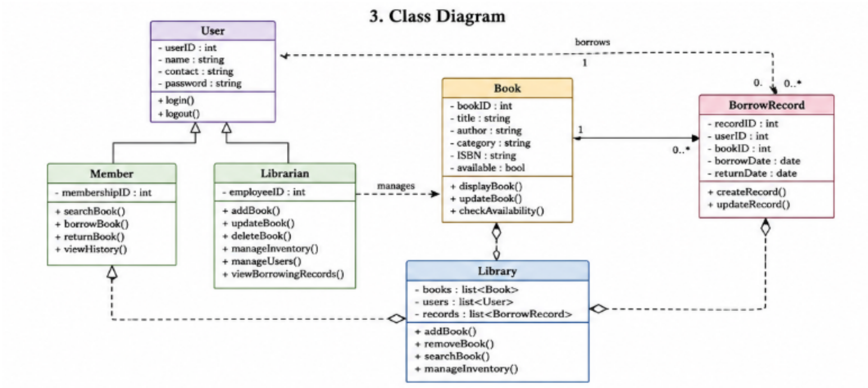

Figure 3: Class Diagram

CONCLUSION

The proposed Library Management System provides an efficient and organized way to manage library resources using C++. The system will allow users to search for books, borrow and return books, while librarians will be able to manage books, users, and library inventory.

The project demonstrates the practical application of important Object-Oriented Programming concepts, including encapsulation, inheritance, polymorphism, and abstraction. UML diagrams will be used to design and visualize the system before implementation, while file handling will provide persistent storage for library records.

Although the initial system will be console-based, it provides a strong foundation for future improvements such as database integration, graphical user interfaces, barcode/RFID systems, online reservations, email notifications, and cloud-based library management.

BIBLIOGRAPHY

- Booch, G., Rumbaugh, J., Jacobson, I. (2005). The unified modeling language user guide (2nd ed.). Addison-Wesley.
- Gamma, E., Helm, R., Johnson, R., Vlissides, J. (1994). Design patterns: Elements of reusable object-oriented software. Addison-Wesley.
- Pressman, R. S., Maxim, B. R. (2020). Software engineering: A practitioner's approach (9th ed.). McGraw-Hill Education.
- Sommerville, I. (2016). Software engineering (10th ed.). Pearson.
- Stroustrup, B. (2013). The C++ programming language (4th ed.). Addison-Wesley.